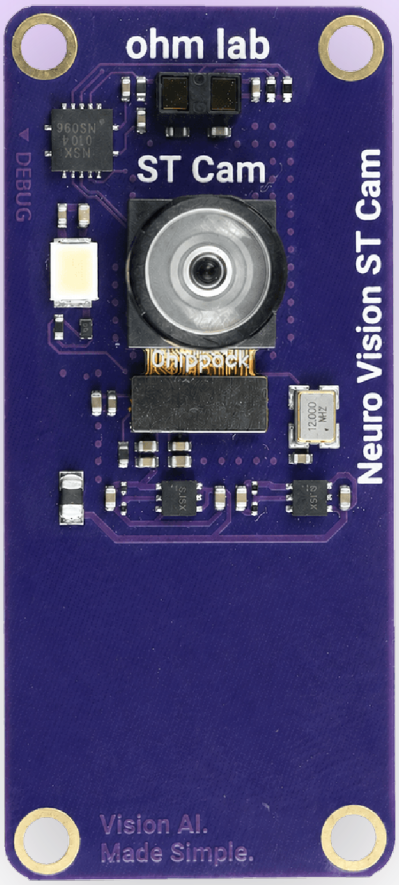
Neuro Vision ST Cam

Datasheet



Camera expansion board for the Neuro N6 that accepts ST camera promodules. It is supplied with the VD66GY 1.5 MP colour global shutter promodule and 152° lens as standard, and adds a VL53L1CB Time-of-Flight ranging sensor and a white illumination LED.

1.5 MP
Standard VD66GY promodule

88 fps
At 640 x 480

152°
Standard lens DFOV

2 lanes
MIPI CSI-2

PART NUMBER	HARDWARE REVISION	DOCUMENT	DATE
OL-DEV-N6-004	Rev B	OL-DEV-N6-004 Rev D	25 October 2026

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1 Overview

The Neuro Vision ST Cam is a camera expansion board for the Neuro N6 that accepts ST camera promodules. It is supplied with the VD66GY global shutter promodule and a 152° lens, which exposes all pixels at the same instant and so removes the motion artefacts of rolling shutter sensors for fast-moving subjects, code reading and machine vision.

- ST camera promodule connector, 30-pin, 0.35 mm pitch.
- Supplied with the CAM-6GY-152VIS promodule: VD66GY 1.5 MP colour global shutter sensor, 152° DFOV lens; up to 1120 x 1364 at 45 fps and 640 x 480 at 88 fps with the Arduino core.
- Promodule options: the VD66GY with a 73° DFOV lens, available now; the VD65G4 0.56 MP colour global shutter sensor, planned as soon as ST promodules are available; the VD1943 5.1 MP RGB-IR sensor, with software support planned.
- 2-lane MIPI CSI-2 video interface at 804 Mbit/s per lane; RAW8 frames demosaiced by the Neuro N6 image signal processor.
- VL53L1CB Time-of-Flight ranging sensor in long distance mode, ranging a 4 x 4 grid of zones.
- White illumination LED switched by the promodule STROBE output, pulsed with each exposure or kept on.
- Powered from the Neuro N6 3.3 V and 1.8 V rails through two on-board LDO regulators.
- One 30-pin and one 40-pin board-to-board connector, 3.5 mm mating height.
- Supported by the Neuro N6 Arduino core.

2 Introduction

The Neuro Vision ST Cam mounts on the Neuro N6 through two board-to-board connectors with a mating height of 3.5 mm. Global shutter operation makes it suitable for fast-moving subjects, barcode and code reading, and machine vision applications.

The camera is an ST camera promodule on a 30-pin, 0.35 mm pitch connector. The board is supplied with the CAM-6GY-152VIS promodule, which carries the VD66GY 1.5 MP colour global shutter sensor behind a 152° DFOV lens. The same sensor is available with a 73° DFOV lens (CAM-6GY-073VIS). Support for the VD65G4 is planned as soon as ST promodules for it are available, and the VD1943 fits the board with software support planned. Resolution, frame rate and field of view depend on the promodule fitted; figures in this datasheet refer to the standard promodule unless stated otherwise.

The Neuro N6 Arduino core, available from neuron6.com, provides drivers and examples for the camera, the Time-of-Flight sensor and the illumination LED.

For detailed information on the ST camera promodules, their image sensors and the VL53L1CB sensor, refer to the STMicroelectronics documentation.

2.1 Camera promodules

OPTION	PROMODULE	SENSOR	LENS	STATUS
Standard	`CAM-6GY-152VIS`	VD66GY, 1.5 MP colour, global shutter	152° DFOV	Fitted as standard
Option	`CAM-6GY-073VIS`	VD66GY, 1.5 MP colour, global shutter	73° DFOV	Available
Planned	VD65G4 promodule	VD65G4, 0.56 MP colour, global shutter	To be confirmed	Planned as soon as ST promodules are available
Future	VD1943 promodule	VD1943, 5.1 MP RGB-IR, global or rolling shutter	Depends on the promodule	Fits the board; software support planned

Table 1. Camera promodule options. Contact Ohm Lab for availability.

2.2 Ordering information

PRODUCT	PART NUMBER	HARDWARE REVISION
Neuro Vision ST Cam	`OL-DEV-N6-004`	Rev B

Table 2. Ordering information. Quote the part number and hardware revision when ordering and when contacting Ohm Lab for support.

2.3 Related products

- Neuro N6 (OL-DEV-N6-001)
- Neuro Vision OV5640 (OL-DEV-N6-002)
- Neuro Vision OV5640-W (OL-DEV-N6-003)
- Neuro Vision Thermal (OL-DEV-N6-005)
- Neuro Vision TFT (OL-DEV-N6-006)
- Neuro Vision TFT-W (OL-DEV-N6-007)
- Neuro N6 Enclosure (OL-DEV-N6-009)
- Neuro N6 Debug Adapter (OL-DEV-N6-010)
- B-CAMS-IMX Adapter (OL-DEV-N6-011)

3 Features

FEATURE	DESCRIPTION
Camera	
Camera promodule	ST camera promodule on a 30-pin, 0.35 mm pitch board-to-board connector. CAM-6GY-152VIS (152° DFOV) fitted as standard; CAM-6GY-073VIS (73° DFOV) available; VD65G4 planned; VD1943 fits, with software support planned
Image sensor (standard promodule)	VD66GY 1.5 MP colour global shutter. With the Arduino core: 320 x 240 and 640 x 480 at 88 fps, 1024 x 768 and 1120 x 720 at 60 fps, 1120 x 1364 at 45 fps; RAW8 output demosaiced by the Neuro N6 image signal processor (fixed, automatic or manual white balance); automatic or manual exposure, gain 1 to 64x; 2-lane MIPI CSI-2 at 804 Mbit/s per lane; I2C control at 1.8 V
Depth sensing	
VL53L1CB Time-of-Flight sensor	Absolute ranging independent of target colour and reflectance; long distance mode with a 4 x 4 grid of zones, one distance per zone; 940 nm Class 1 VCSEL; 27° field of view; interrupt routed to CONN_E (the Arduino core polls the sensor over I2C)
Illumination	
White LED	2835-package white LED switched by a transistor from the promodule STROBE output; pulsed with each exposure or kept on, selected at start-up
Interfaces	
Board-to-board connectors	40-pin and 30-pin HC-PBB40C receptacles, 0.4 mm pitch, 3.5 mm mating height: 2-lane MIPI CSI-2, I2C (3.3 V host side), Time-of-Flight interrupt, NRST
Power	
Supply	3.3 V and 1.8 V from the Neuro N6; two on-board LDO regulators (2.8 V analog shared with the Time-of-Flight sensor, 1.15 V camera core); ferrite-filtered supply input

Table 3. Neuro Vision ST Cam features.

4 Mechanical

The board follows the Adafruit Feather outline with the four mounting holes on the 45.72 x 17.78 mm pattern. All dimensions are in millimetres and are taken from the PCB design data and the 3D model. The board mounts back to back with the Neuro N6, so the camera side faces away from the Neuro N6.

PCB outline	50.80 x 22.86 mm, corner radius 2.54 mm
PCB	1.6 mm
Height above the PCB	3.3 mm (promodule as in the 3D model; the height depends on the promodule and lens fitted)
Height below the PCB	3.4 mm (board-to-board receptacles)
Overall height	8.3 mm
Stack height on the Neuro N6	15.5 mm from the Neuro N6 battery connector to the top of the expansion board, with a 3.5 mm mating gap
Mounting holes	4 x Ø2.54 mm plated, 45.72 x 17.78 mm pattern, 2.54 mm from the board edges, connected to GND
Board-to-board receptacles	40-pin centred 17.61 mm and 30-pin centred 12.37 mm from the Feather body edge of the Neuro N6 when mated; both on the board centre line

Table 4. Mechanical summary.

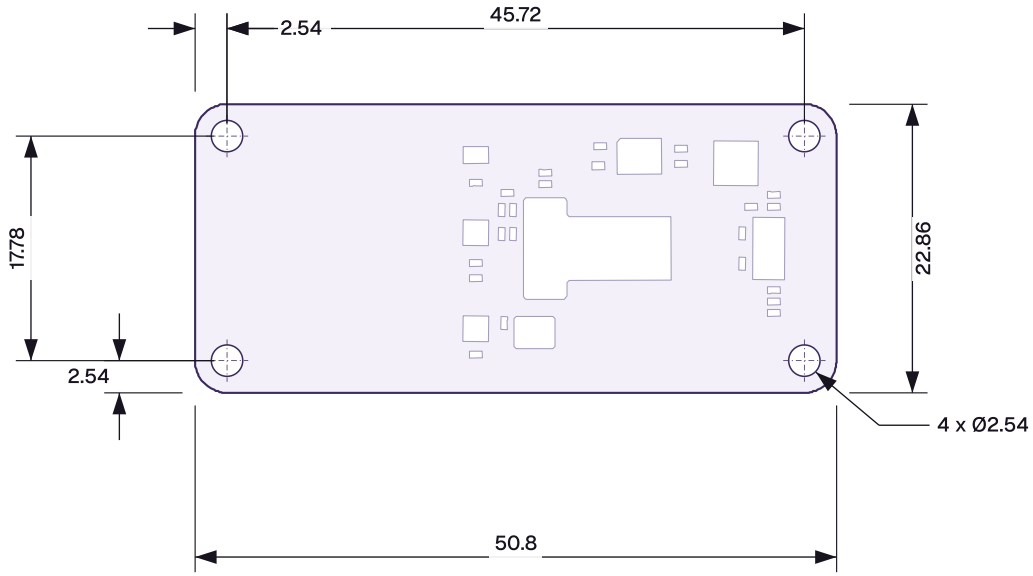


Figure 1. Neuro Vision ST Cam top view (component side). Component outlines are taken from the 3D model; dimensions are nominal design values in millimetres.

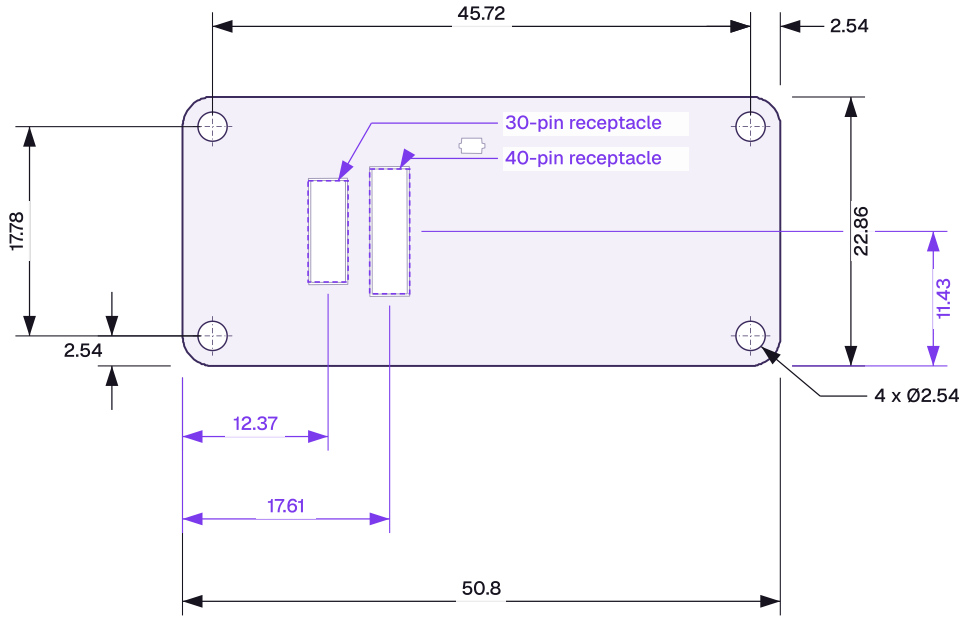


Figure 2. Neuro Vision ST Cam bottom view (mirrored, as seen from below), showing the board-to-board receptacles. Positions match the plug positions on the Neuro N6.

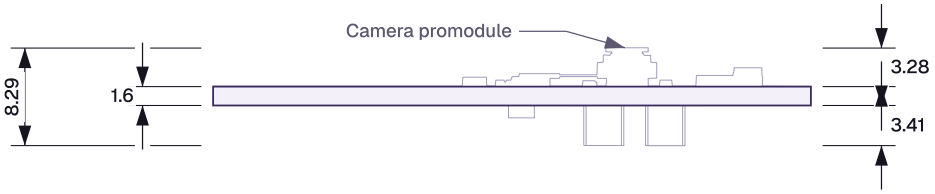


Figure 3. Neuro Vision ST Cam side view with a promodule fitted, as in the 3D model.

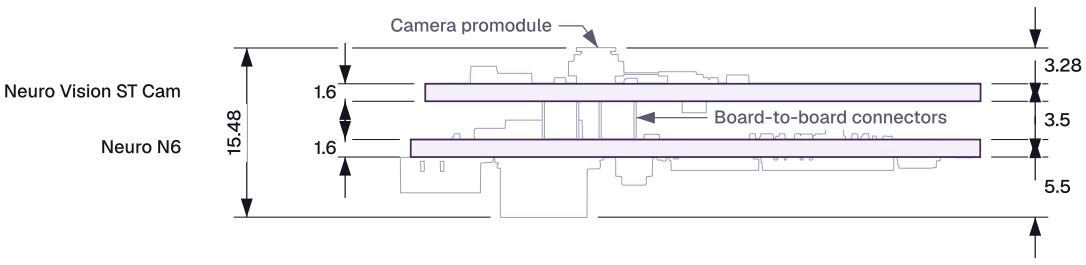


Figure 4. Neuro Vision ST Cam mounted on the Neuro N6, side view to scale from the two 3D models. The boards mount back to back with a 3.5 mm mating gap; the Neuro N6 component side faces down.

5 Block diagram

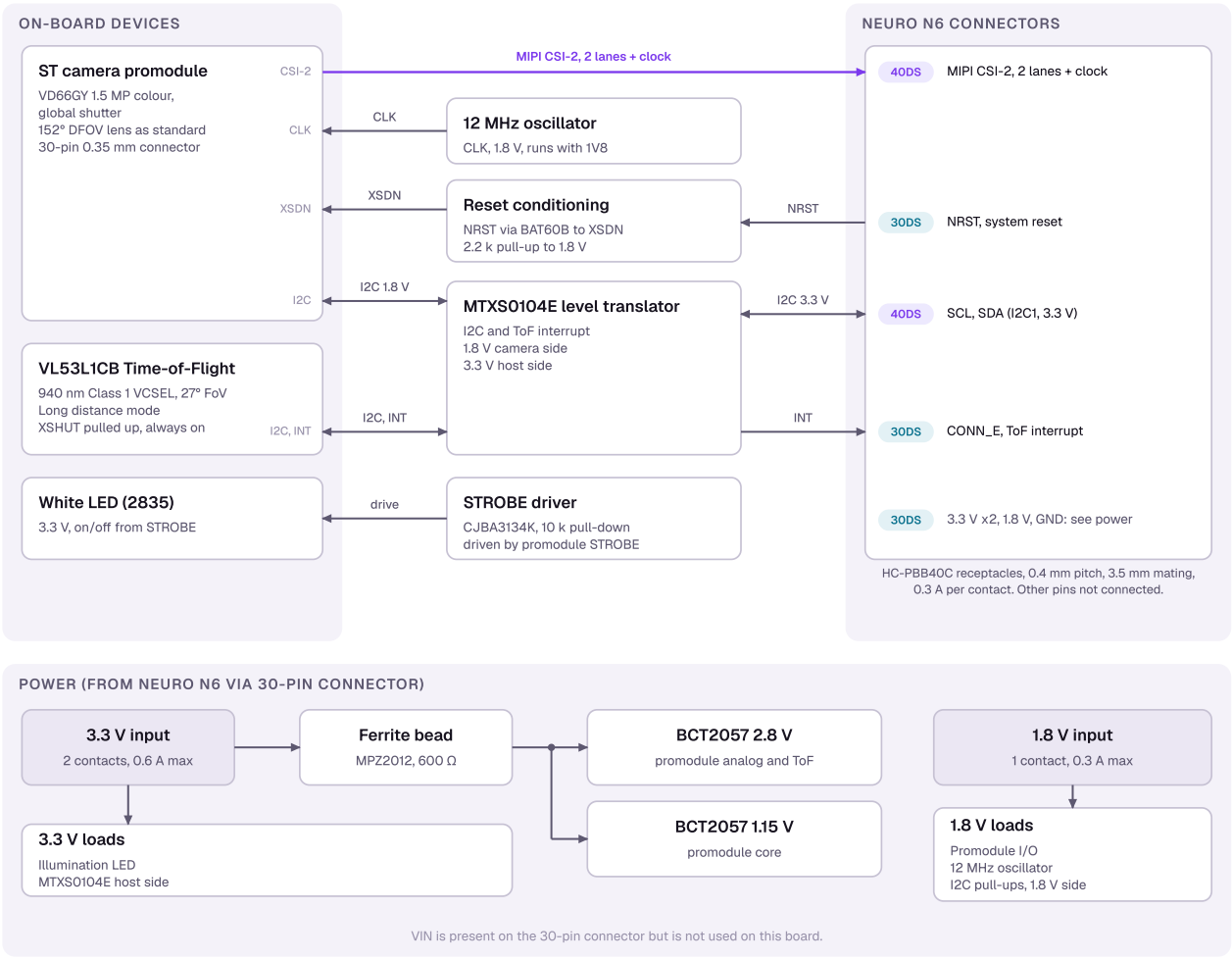


Figure 5. Neuro Vision ST Cam block diagram.

6 Hardware interfaces

6.1 Camera module

The camera is an ST camera promodule on a 30-pin, 0.35 mm pitch board-to-board connector. The board is supplied with the CAM-6GY-152VIS, which carries the VD66GY colour global shutter sensor and a 152° DFOV lens. The sensor outputs RAW8 frames on the 2-lane MIPI CSI-2 interface routed to the 40-pin connector, which the Neuro N6 image signal processor demosaics. It is clocked by an on-board 12 MHz oscillator and is shut down by the system NRST signal (see the reset and clocking section). The promodule spare GPIO signals are not connected on this board.

> **Caution** The camera module, its connector and the board are sensitive to electrostatic discharge and mechanical stress. Handle the board by its edges and do not connect or disconnect the camera module while the board is powered.

6.2 Time-of-Flight sensor

A VL53L1CB ranging sensor is fitted next to the camera module. It emits invisible 940 nm light from an integrated Class 1 VCSEL and measures the photon return time on a SPAD array, giving absolute distance independent of target colour and reflectance over a 27° field of view. The Arduino core runs the sensor in long distance mode and ranges a 4 x 4 grid of zones, one distance per zone. The sensor shares the 1.8 V I2C bus with the camera. Its interrupt output (GPIO1) is level-translated to 3.3 V and routed to CONN_E (30-pin connector pin 9); the Arduino core polls the sensor over I2C and does not use the interrupt. XSHUT is pulled up to 1.8 V on the board, so the sensor is enabled whenever the board is powered. See the laser safety section for the classification of the emitter and the handling precautions.

6.3 Illumination LED

A high-brightness white LED provides scene illumination. It is powered from the 3.3 V rail and switched by a CJBA3134K transistor driven from the promodule STROBE output; a 10 kΩ pull-down holds the LED off while the camera is shut down. STROBE is configured over I2C: the Arduino core pulses the LED with each exposure or keeps it on, selected at start-up. No series resistor is fitted: the LED current is set by the LED forward voltage, the 3.3 V rail and the transistor on-resistance. Take this into account before leaving the LED on for long periods.

6.4 Connectors

CONNECTOR	PART NUMBER	MATING PART
Camera module connector	`BM28B0.6-30DS/2-0.35V(51)`	`BM28B0.6-30DP/2-0.35V(51)` (on the module)
30-pin board-to-board receptacle	`HC-PBB40C-30DS-0.4V-3.5-02`	`HC-PBB40C-30DP-0.4V-02` (Neuro N6)
40-pin board-to-board receptacle	`HC-PBB40C-40DS-0.4V-3.5-02`	`HC-PBB40C-40DP-0.4V-02` (Neuro N6)

Table 5. Connector part numbers.

The board-to-board receptacles mate with the plugs fitted to the underside of the Neuro N6 at a mating height of 3.5 mm. The two boards mount back to back, so the component sides of both boards face outwards. Align both connectors before pressing the boards together, and separate the boards evenly when removing the expansion board.

7 Board operation

7.1 Absolute maximum ratings

Absolute maximum ratings are stress ratings only. They are not operating conditions, and functional operation at or beyond these limits is neither implied nor guaranteed. Exposure to conditions at the limits may reduce reliability. Each limit is the lowest rating among the components connected to that net on the Neuro Vision ST Cam schematic. All voltages are referenced to GND.

PARAMETER	MIN	MAX	UNIT	LIMITING ELEMENT
Supply voltage, VDD_3V3	-0.3	6.0	V	MTXS0104E supply (VCCB)
Supply voltage, VDD_1V8	-0.3		V	Camera promodule I/O supply: refer to the ST promodule datasheet (VL53L1CB I/O 3.6 V)
Voltage on host-side 3.3 V signals (SCL, SDA, CONN_E)	-0.3	VDD_3V3 + 0.3	V	MTXS0104E B port
Voltage on NRST	-0.3	VDD_3V3 + 0.3	V	Neuro N6 reset line at 3.3 V; BAT60B on this board
Voltage on camera-domain 1.8 V signals (SCL1V8, SDA1V8)	-0.3	VDD_1V8 + 0.3	V	MTXS0104E A port, VL53L1CB, promodule
Continuous current per board-to-board contact		0.3	A	HC-PBB40C 0.4 mm pitch contact
Illumination LED forward current		60	mA	XL-2835UWC-02
Storage temperature	-40	85	°C	Component ratings

Table 6. Absolute maximum ratings.

The 3.3 V rail reaches the board on two contacts of the 30-pin connector and the 1.8 V rail on one, giving 0.6 A and 0.3 A respectively at the 0.3 A per-contact limit. The VIN pins on the 30-pin connector are not connected on this board.

7.2 Recommended operating conditions

PARAMETER	MIN	MAX	UNIT	CONDITIONS
Ambient temperature	0	30	°C	Indoor, dry locations only
Relative humidity	25	75	%	Non-condensing
Supply				From the Neuro N6, powered with 5 V DC via USB-C or its battery

Table 7. Recommended operating conditions.

7.3 Electrical characteristics

PARAMETER	MIN	TYP	MAX	UNIT	CONDITIONS
Supply voltage, VDD_3V3	3.23	3.33	3.44	V	From the Neuro N6
Supply voltage, VDD_1V8	1.75	1.8	1.85	V	From the Neuro N6
On-board analog rail, 2V8		2.8		V	BCT2057, camera analog and Time-of-Flight
On-board camera core rail, 1V15		1.15		V	BCT2057
Camera-domain logic level		1.8		V	I2C, clock, shutdown
Host-side logic level		3.3		V	I2C, interrupt

Table 8. Electrical characteristics.

7.4 Power supply

The board has no power input of its own. It is powered from the Neuro N6 through the board-to-board connectors, which provide the 3.3 V (VDD_3V3) and 1.8 V (VDD_1V8) rails. The 3.3 V rail is filtered by a ferrite bead and feeds two BCT2057 low-dropout regulators: one generates the 2.8 V analog supply shared by the camera module and the Time-of-Flight sensor with its VCSEL, and one generates the 1.15 V camera core supply. The camera I/O domain, the 12 MHz oscillator and the I2C pull-ups run directly from the 1.8 V rail.

RAIL	SOURCE	VOLTAGE	SUPPLIES
`VDD_3V3`	Neuro N6, 30-pin connector	3.3 V	Illumination LED, I2C level translator, LDO inputs (via ferrite bead)
`VDD_1V8`	Neuro N6, 30-pin connector	1.8 V	Camera I/O, 12 MHz oscillator, I2C pull-ups
`2V8`	On-board LDO	2.8 V	Camera analog supply, Time-of-Flight sensor and VCSEL
`1V15`	On-board LDO	1.15 V	Camera digital core
`VIN`	Neuro N6, 30-pin connector	VBUS or VBAT	Not used on this board

Table 9. Power rails.

> **Warning** Do not attach or detach the Neuro Vision ST Cam while the Neuro N6 is powered. Disconnect the USB-C supply and the battery before changing expansion boards; otherwise the boards may be permanently damaged.

7.5 Reset and clocking

The system reset signal NRST from the Neuro N6 controls the camera: the sensor's active-low XSDN (shutdown) input is pulled up to the 1.8 V rail through 2.2 kΩ and is pulled low through a BAT60B diode whenever NRST is asserted, so the camera is shut down during a system reset and released afterwards. The Time-of-Flight sensor is not connected to NRST; it is reset over I2C or by removing board power. The 12 MHz oscillator runs whenever the 1.8 V rail is present and drives the module clock input directly.

8 Peripheral interfaces

Connector pins are given as `40DS:n` and `30DS:n`, the receptacle pin numbers on this board. The matching Neuro N6 plug pin is listed in the connector tables. The Neuro N6 datasheet gives the MCU pin for every signal.

8.1 MIPI CSI-2

The promodule 2-lane MIPI CSI-2 interface is routed differentially from the promodule connector to the 40-pin connector, matching the Neuro N6 D-PHY receiver (CSI-2 version 1.3 compatible). The Arduino core runs the VD66GY at 804 Mbit/s per lane. Ground guard pins separate the lanes.

NET	CONNECTOR PIN	DESCRIPTION
`CSI_D0P`	`40DS:31`	Data lane 0 positive
`CSI_D0N`	`40DS:29`	Data lane 0 negative
`CSI_CLKP`	`40DS:23`	Clock lane positive
`CSI_CLKN`	`40DS:21`	Clock lane negative
`CSI_D1P`	`40DS:15`	Data lane 1 positive
`CSI_D1N`	`40DS:13`	Data lane 1 negative

Table 10. MIPI CSI-2 pins.

8.2 I2C

A single I2C bus controls the camera and the Time-of-Flight sensor. It enters the board at 3.3 V on the 40-pin connector, from the Neuro N6 I2C1 bus that also serves the on-board IMU and magnetometer of the Neuro N6, and is level-translated by an MTXS0104E to the 1.8 V camera domain, where it is pulled up through 2.2 kΩ resistors. Standard mode (100 kHz) and fast mode (400 kHz) are supported.

NET	LOGIC LEVEL	CONNECTOR PIN	DESCRIPTION
`SCL`	3.3 V	`40DS:35`	I2C clock, host side
`SDA`	3.3 V	`40DS:37`	I2C data, host side
`SCL1V8`	1.8 V		I2C clock, camera domain
`SDA1V8`	1.8 V		I2C data, camera domain

Table 11. I2C pins.

DEVICE	7-BIT ADDRESS	NOTES
Promodule image sensor	`0x10`	VD66GY on the standard promodule; refer to STMicroelectronics for other promodules
VL53L1CB Time-of-Flight sensor	`0x29`	Default address, can be reprogrammed at run time

Table 12. I2C addresses.

8.3 Camera control signals

NET	PROMODULE SIGNAL	CONNECTOR PIN	DESCRIPTION
NRST	XSDN	30DS:29	Active-low camera shutdown, driven by the system NRST (2.2 kΩ pull-up to 1.8 V)
CLK	CLK		12 MHz master clock from the on-board oscillator
STROBE	STROBE		Strobe output, switches the white LED
GPIO1..5			Promodule GPIO, not connected

Table 13. Camera control signals.

8.4 Time-of-Flight signals

NET	VL53L1CB PIN	CONNECTOR PIN	DESCRIPTION
CONN_E	GPIO1	30DS:9	Ranging interrupt at 3.3 V, level translated; not used by the Arduino core, which polls the sensor
	XSHUT		Pulled up to 1.8 V on the board; the sensor is always enabled

Table 14. Time-of-Flight signals.

8.5 40-pin board-to-board connector

PIN	NET	N6 PLUG PIN	FUNCTION
1	SD_D1	40	SDMMC (not used on this board)
2	LCD_R3	39	LCD interface (not used on this board)
3	SD_CMD	38	SDMMC (not used on this board)
4	LCD_R7	37	LCD interface (not used on this board)
5	SD_CK	36	SDMMC (not used on this board)
6	LCD_R6	35	LCD interface (not used on this board)
7	SD_D3	34	SDMMC (not used on this board)
8	SD_D0	33	SDMMC (not used on this board)
9	GND	32	Ground
10	LCD_VSYNC	31	LCD interface (not used on this board)
11	GND	30	Ground
12	CONN_G	29	GPIO (not used on this board)
13	CSI_D1N	28	MIPI CSI-2 data lane 1 negative
14	LCD_CLK	27	LCD interface (not used on this board)
15	CSI_D1P	26	MIPI CSI-2 data lane 1 positive
16	LCD_B5	25	LCD interface (not used on this board)
17	GND	24	Ground
18	LCD_B3	23	LCD interface (not used on this board)

PIN	NET	N6 PLUG PIN	FUNCTION
19	GND	22	Ground
20	LCD_B4	21	LCD interface (not used on this board)
21	CSI_CLKN	20	MIPI CSI-2 clock lane negative
22	LCD_B6	19	LCD interface (not used on this board)
23	CSI_CLKP	18	MIPI CSI-2 clock lane positive
24	LCD_B7	17	LCD interface (not used on this board)
25	GND	16	Ground
26	CONN_F	15	GPIO (not used on this board)
27	GND	14	Ground
28	LCD_G5	13	LCD interface (not used on this board)
29	CSI_D0N	12	MIPI CSI-2 data lane 0 negative
30	LCD_G3	11	LCD interface (not used on this board)
31	CSI_D0P	10	MIPI CSI-2 data lane 0 positive
32	LCD_G7	9	LCD interface (not used on this board)
33	GND	8	Ground
34	LCD_G6	7	LCD interface (not used on this board)
35	SCL	6	I2C clock, 3.3 V host side (Neuro N6 I2C1)
36	LCD_R4	5	LCD interface (not used on this board)
37	SDA	4	I2C data, 3.3 V host side (Neuro N6 I2C1)
38	CONN_C	3	GPIO (not used on this board)
39	SD_D2	2	SDMMC (not used on this board)
40	CONN_D	1	GPIO (not used on this board)

Table 15. 40-pin receptacle pin assignment.

8.6 30-pin board-to-board connector

PIN	NET	N6 PLUG PIN	FUNCTION
1	LCD_HSYNC	29	LCD interface (not used on this board)
2	SAI1_MCLK_B	30	Audio interface (not used on this board)
3	CONN_J	27	GPIO (not used on this board)
4	LCD_R5	28	LCD interface (not used on this board)
5	CONN_H	25	GPIO (not used on this board)
6	CONN_N	26	GPIO (not used on this board)
7	CONN_M	23	GPIO (not used on this board)
8	CONN_L	24	GPIO (not used on this board)
9	CONN_E	21	Time-of-Flight interrupt to the host
10	DEBUG_CLK	22	SWD clock (not used on this board)
11	LCD_DE	19	LCD interface (not used on this board)
12	DEBUG_DAT	20	SWD data (not used on this board)
13	VIN	17	VIN rail from the Neuro N6 (not used on this board)
14	LCD_G4	18	LCD interface (not used on this board)
15	VIN	15	VIN rail from the Neuro N6 (not used on this board)
16	SAI1_SCK_B	16	Audio interface (not used on this board)
17	VIN	13	VIN rail from the Neuro N6 (not used on this board)
18	SAI1_SD_B	14	Audio interface (not used on this board)
19	VIN	11	VIN rail from the Neuro N6 (not used on this board)
20	CONN_K	12	GPIO (not used on this board)
21	VIN	9	VIN rail from the Neuro N6 (not used on this board)
22	CONN_B	10	GPIO (not used on this board)
23	3V3	7	3.3 V supply input
24	GND	8	Ground
25	3V3	5	3.3 V supply input
26	LCD_G2	6	LCD interface (not used on this board)
27	1V8	3	1.8 V supply input
28	SAI1_FS_B	4	Audio interface (not used on this board)
29	NRST	1	System reset input (shuts down the camera)
30	CONN_A	2	GPIO (not used on this board)

Table 16. 30-pin receptacle pin assignment.

9 Laser safety

The Neuro Vision ST Cam contains a laser emitter. The VL53L1CB Time-of-Flight sensor uses an integrated vertical-cavity surface-emitting laser (VCSEL) and emits invisible 940 nm infrared light through the optical window on the top face of the sensor package while a ranging measurement is in progress. No other component on the board emits laser radiation.

9.1 Laser classification

The VL53L1CB is a Class 1 laser product and is safe under all conditions of reasonably foreseeable normal use. No protective eyewear or other protective measures are required, provided the board is used as described in this document.

Laser source	VCSEL integrated within the VL53L1CB
Laser class	Class 1
Classification standard	IEC 60825-1:2014 (Edition 3.0)
Wavelength	940 nm, invisible infrared
Field of illumination	27° full field of view
Aperture	Optical window on the top face of the VL53L1CB package



Figure 6. Class 1 laser product label, as printed on the leaflet supplied with the board.

9.2 Handling precautions

- The emitted light is invisible. Do not assume the emitter is inactive because no light can be seen: the VCSEL is enabled whenever the host has started a ranging measurement.
- Do not examine the sensor window at close range through magnifying optics such as an eye loupe, a magnifier or a microscope.
- Do not fit additional lenses, mirrors or other optical elements in front of the sensor. External optics can concentrate the beam and invalidate the Class 1 classification.
- Do not operate a board on which the VL53L1CB package or its optical window is cracked, scratched, contaminated or otherwise damaged.
- Do not rework, remove or replace the VL53L1CB. The sensor contains no user-serviceable parts, and disassembly may expose the emitter without its optical window.
- Configure the sensor through the ST VL53L1CB API or the Neuro N6 Arduino core drivers, and keep ranging parameters within the ranges given in the ST documentation.

The white illumination LED is not a laser product and falls outside the scope of IEC 60825-1. It is nevertheless a high-brightness source: avoid looking directly into the LED at close range while it is switched on.

Where the Neuro Vision ST Cam is built into an end product, the manufacturer of that product is responsible for its compliance with IEC 60825-1, including any labelling and user information.

10 Regulatory compliance

This section describes the regulatory status of the Neuro Vision ST Cam as supplied by Ohm Lab. The board is an expansion board intended to be operated only when mounted on a Neuro N6, and it is assessed as part of that combination rather than as a stand-alone product. Where the board is incorporated into a finished product, responsibility for the compliance of that product rests with its manufacturer.

10.1 Product marking

SYMBOL	MARKING	APPLIED TO	MEANING
CE	CE	Board and packaging	Conformity with the EMC Directive 2014/30/EU and the RoHS Directive 2011/65/EU.
	Crossed-out wheeled bin	Board and packaging	Do not dispose of with unsorted household waste. The bar identifies equipment placed on the market after 13 August 2005.
	Triman and Info-tri	Packaging	France: sorting information required by the French environment code. The Triman is printed on the carton; an Info-tri sticker with the sorting instructions is added to orders for France.
	Class 1 laser product	Leaflet	Explanatory label required by IEC 60825-1:2014 for the VL53L1CB emitter, printed on the leaflet supplied with the board. See the laser safety section.

Table 17. Product markings. The controlled artwork is held in the technical file.

No UKCA marking is applied: CE marking is accepted for goods placed on the Great Britain market. RoHS conformity is declared through the CE marking and carries no separate mark.

10.2 Electromagnetic compatibility

The Neuro N6 platform, including the Neuro Vision ST Cam, has been assessed for electromagnetic compatibility as a product family, based on tests of worst-case configurations of the Neuro N6 with expansion boards fitted. The Neuro Vision ST Cam conforms to the EMC Directive 2014/30/EU, with emissions assessed to EN 55032 Class B and immunity to EN 55035. The assessment applies to the board mounted on a Neuro N6 and operated with the default clock settings of the Neuro N6 Arduino core. The declaration of conformity is available from Ohm Lab on request.

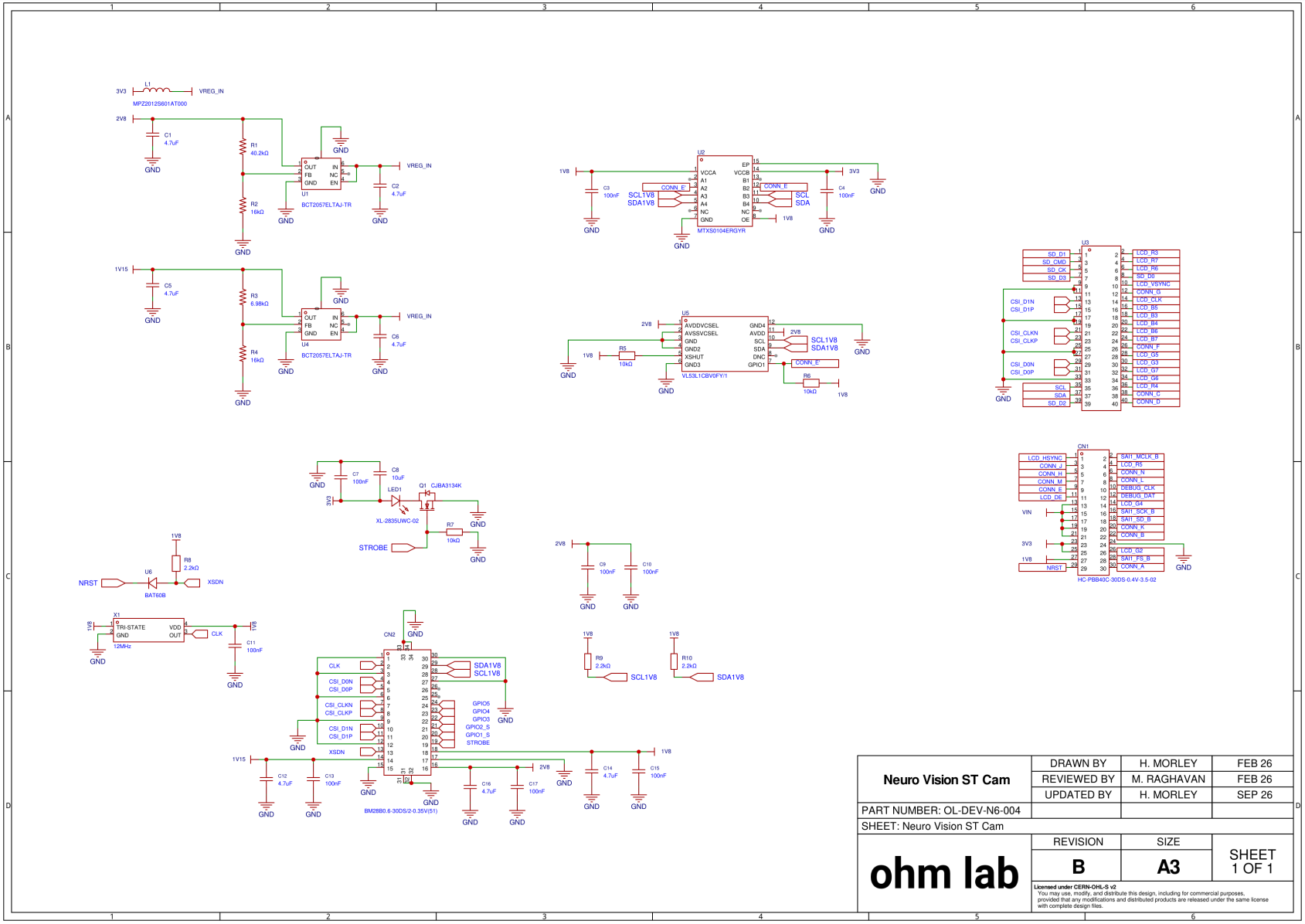
10.3 Materials and disposal

The Neuro Vision ST Cam is manufactured to the substance restrictions of Directive 2011/65/EU (RoHS 2) as amended and the UK Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment Regulations 2012. Supplier material declarations, and declarations covering substances of very high concern under REACH, are held in the technical file.

At end of life, do not dispose of the board with unsorted household waste. Return it to a designated collection point for electrical and electronic equipment, or to Ohm Lab.

11 Schematics

The schematics are reproduced on the following pages for reference. Net names match those used in this document. Full-resolution sheets are published in the Neuro N6 Hardware repository (github.com/ohmlab-ltd/Neuro-N6-Hardware).



Neuro Vision ST Cam Rev B schematic: camera, Time-of-Flight sensor, illumination, power and connectors. XSDN pull-up R8 shown at 2.2 kΩ, as fitted.

12 Document revision history

REVISION	DATE	CHANGES
A	02/07/2026	• Pre-production first release
B	02/09/2026	• Ratings, laser safety and regulatory section added
C	24/09/2026	• Rev B hardware • New block diagram and drawings
D	25/10/2026	• Features as supported by the Arduino core • XSDN pull-up 2.2 kΩ

Table 18. Document revision history.